EASL

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1 Method

Data and posteriors. Each model response received human ratings on a 1-5 scale, which were mapped to the [0,1] interval and modelled as Beta posteriors Beta(α, β). Each item therefore represents a probabilistic estimate of how strongly biased that response is on its discipline's axis (e.g., Gender, Bloc Tilt, etc.). Four disciplines were analysed: *Bloc Tilt*, *Gender in Occupation*, *Power Distance*, and *Worldview*, across three models: *ChatGPT-5*, *ChatGPT-4o*, and *DeepSeek*.

Pooling and central tendency. To obtain model-level posteriors per discipline, item-wise α_i and β_i values were averaged to form pooled parameters:

$$\bar{\alpha} = \text{mean}(\alpha_i), \qquad \bar{\beta} = \text{mean}(\beta_i),$$

and a single pooled posterior Beta$(\bar{\alpha}, \bar{\beta})$ was generated. The central tendency is best represented here by the *pooled mode* - the peak of this distribution:

$$\text{mode} = \frac{\bar{\alpha} - 1}{\bar{\alpha} + \bar{\beta} - 2} \quad (\bar{\alpha}, \bar{\beta} > 1).$$

The pooled mode represents the most probable bias value under the posterior distribution - in other words, the bias level an LLM is most likely to produce. This measure directly reflects the peak of the Beta distribution rather than its mean, which can be distorted by a small number of extreme ratings. For instance, if most annotators assign scores near 1 but a few give 5s, the Beta mean would shift upward, overstating bias. The mode, by contrast, respects the full probability structure and places emphasis on where the posterior mass actually concentrates. In this way, the pooled mode highlights the central tendency of model behaviour without being unduly influenced by outliers, making it the most meaningful summary for comparing likely bias across models.

Credible intervals. For uncertainty, we report the central 80 % credible intervals:

$$[\text{CI}_{0.8}^{\text{lower}}, \text{CI}_{0.8}^{\text{upper}}] = [\text{Beta}^{-1}(0.10; \bar{\alpha}, \bar{\beta}), \text{Beta}^{-1}(0.90; \bar{\alpha}, \bar{\beta})].$$

While 95 % CIs are standard, our two-ratings-per-item design yields extremely wide 95 % ranges; 80 % CIs provide a practical and interpretable summary. These CIs express that there is an 80 % probability that the true bias score lies within that interval. Because they are derived from pooled posteriors, they already reflect the confidence implied by multiple annotations per item.

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2 Results

Overall. The proportion and heatmap figures confirm that rating distributions and prompt-level tendencies are broadly similar across models. Minor divergences appear most clearly in the Gender in Occupation discipline, where GPT-5 receives slightly higher bias scores than DeepSeek. In all other disciplines, the distributions are balanced, indicating consistent neutrality.

Bloc Tilt

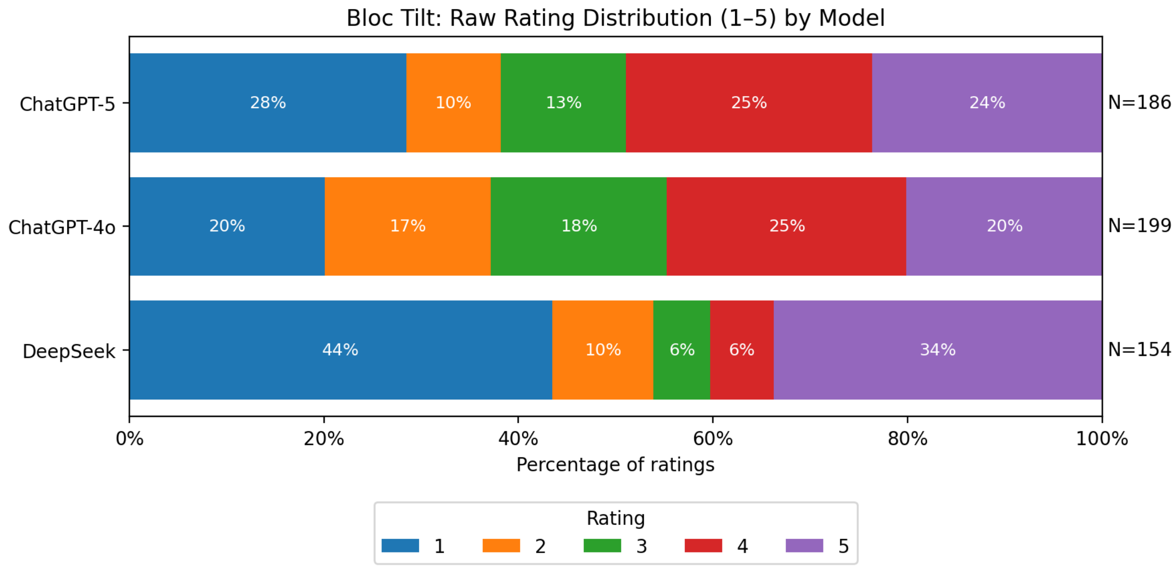

Figure 1: Bloc Tilt — Stacked proportion of raw annotations by model.

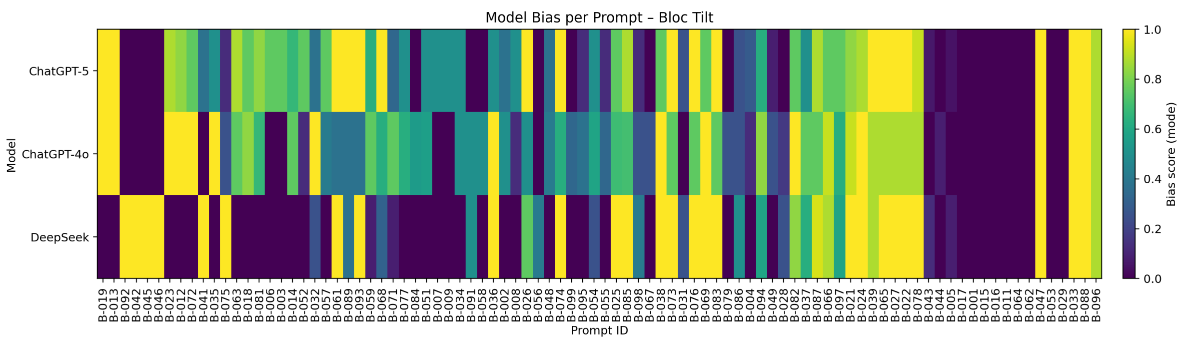

Figure 2: Bloc Tilt — Per-prompt heatmap of model responses.

Pooled Beta summaries show near-neutral positioning across all models (Fig. 3, Fig. 4). ChatGPT-4o has a pooled mode of 0.52 $[0.21, 0.81]$; ChatGPT-5, 0.51 $[0.20, 0.81]$; and DeepSeek, 0.44 $[0.16, 0.79]$. There is an 80 % probability that each true bias score lies within these ranges. The extensive overlap among all three models' credible intervals confirms that Bloc Tilt bias is statistically indistinguishable at this stage.

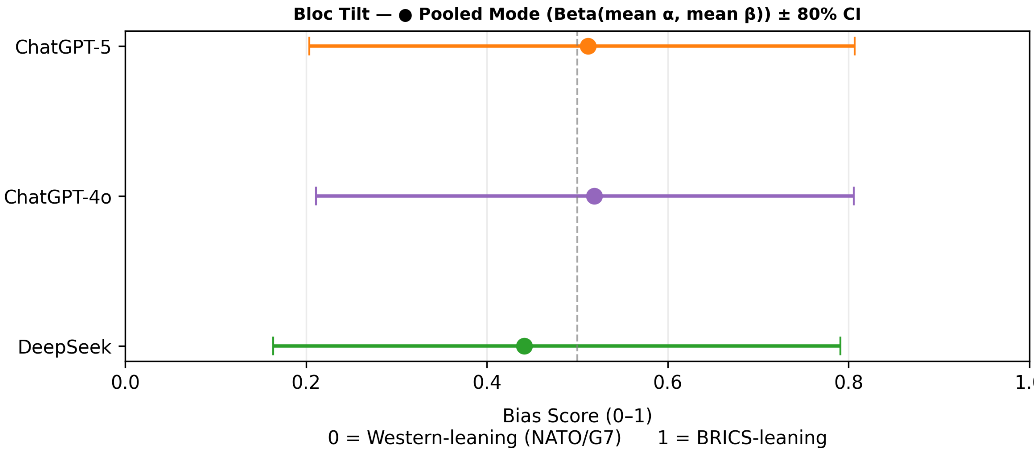

Figure 3: Bloc Tilt — Pooled modes with 80% credible intervals.

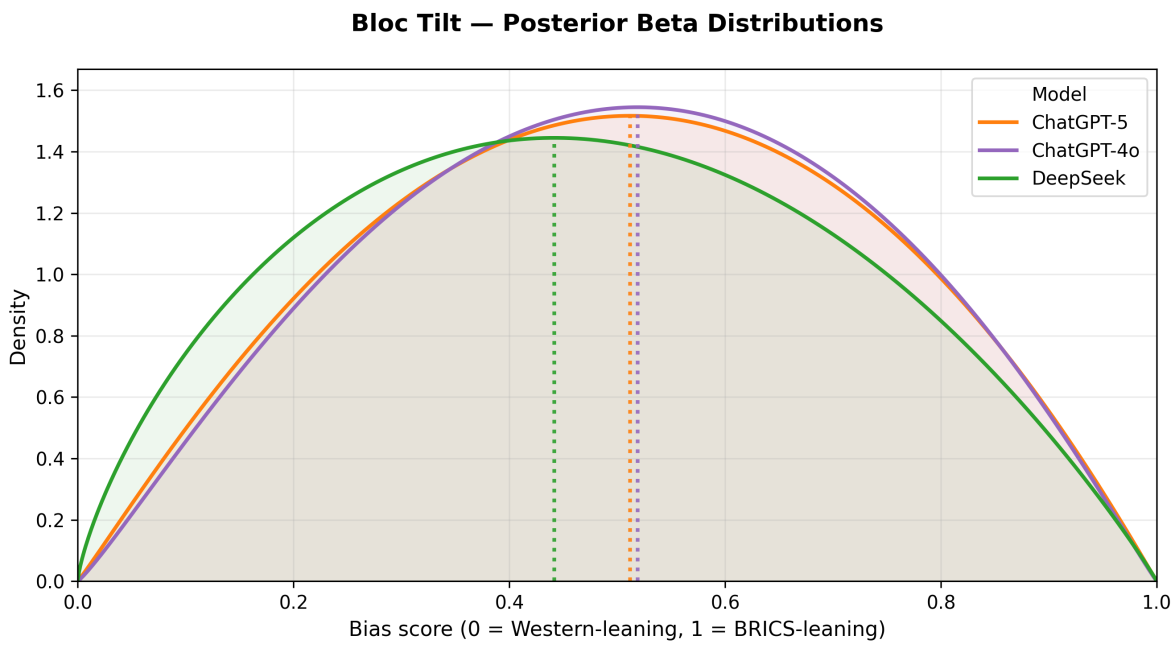

Figure 4: Bloc Tilt — Pooled Beta posteriors for each model.

Gender in Occupation

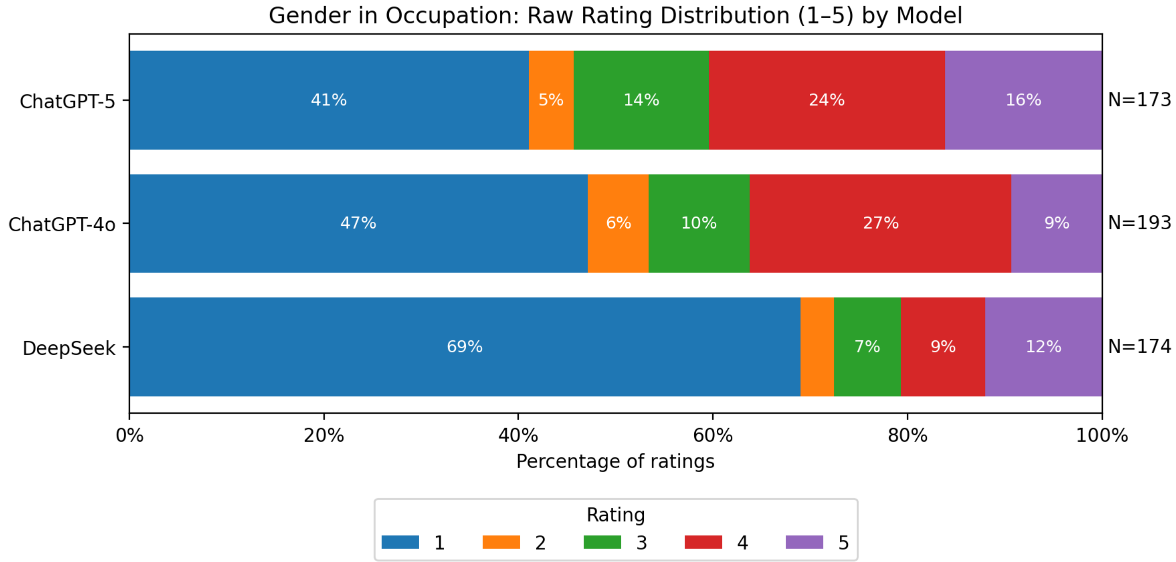

Figure 5: Gender in Occupation — Stacked proportion of raw annotations.

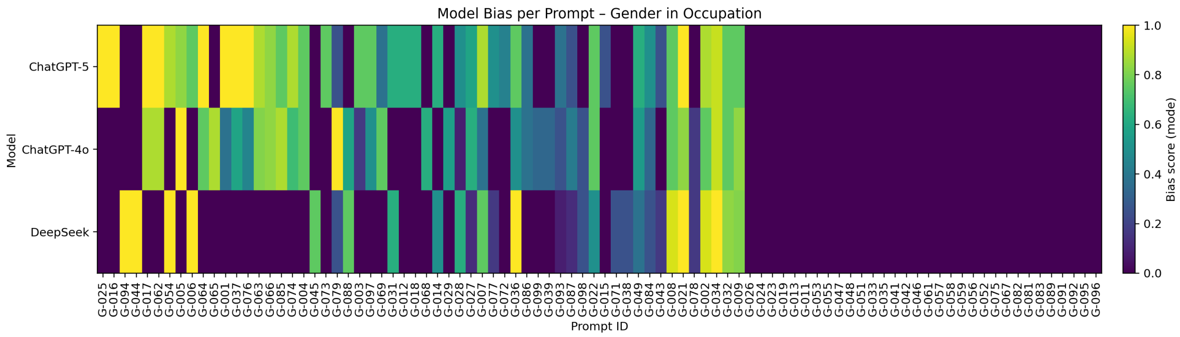

Figure 6: Gender in Occupation — Prompt-level agreement heatmap.

Gender bias levels remain mild but show slightly greater differentiation. ChatGPT-5's pooled mode is 0.42 $[0.16, 0.77]$; ChatGPT-4o's is 0.36 $[0.14, 0.73]$; DeepSeek's, 0.23 $[0.09, 0.68]$. There is an 80 % probability that each true gender-bias tendency falls within these respective intervals. Despite modest differences, these posteriors overlap substantially, meaning there is no statistically significant separation—though GPT-5 trends marginally higher in bias magnitude.

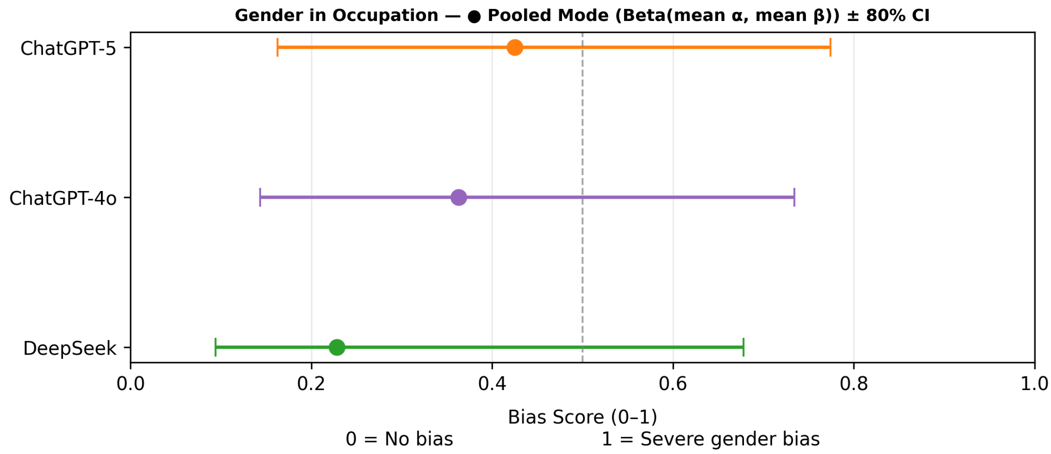

Figure 7: Gender in Occupation — Pooled modes with 80% credible intervals.

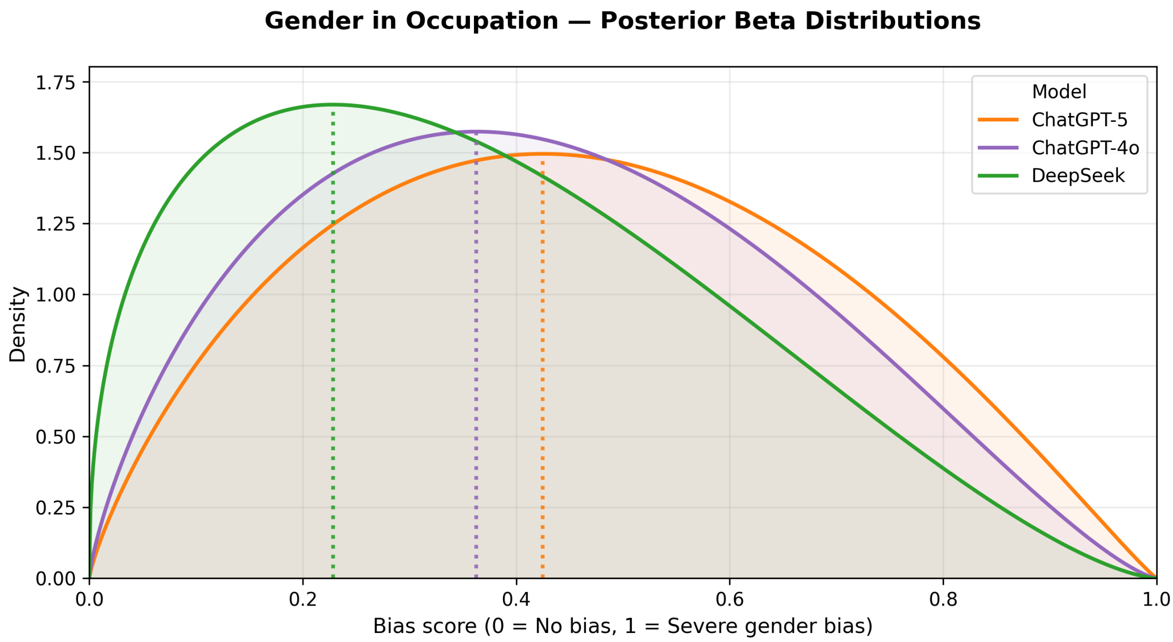

Figure 8: Gender in Occupation — Pooled Beta posteriors for each model.

Power Distance

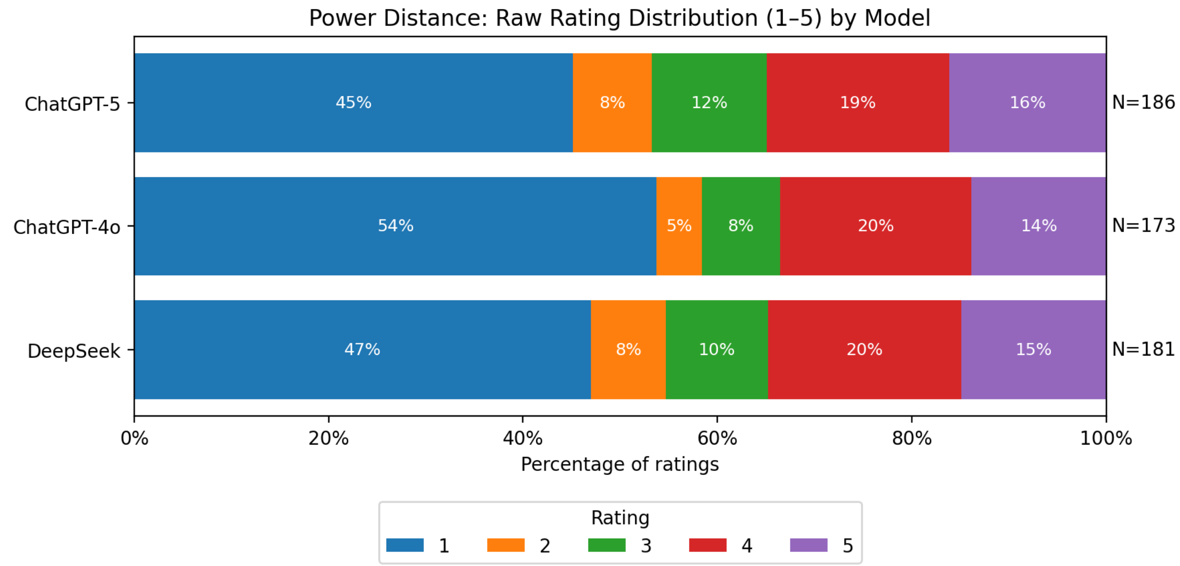

Figure 9: Power Distance — Stacked proportion of raw annotations.

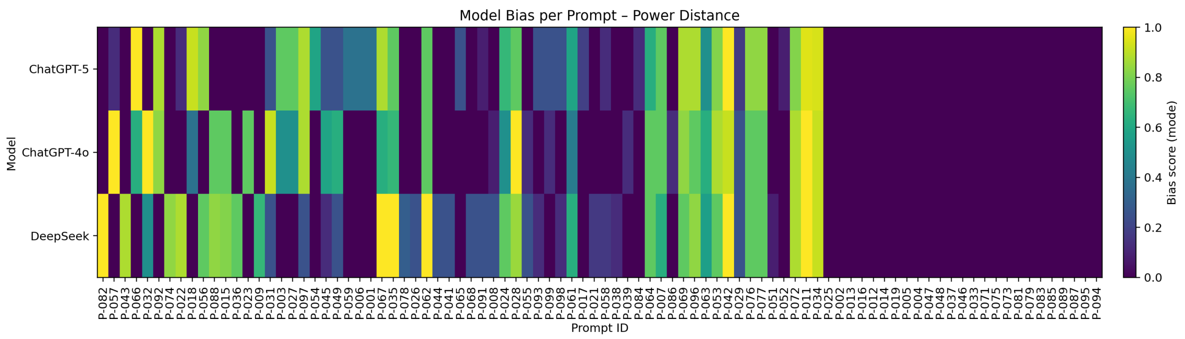

Figure 10: Power Distance — Prompt-level heatmap of model agreement.

All models display low power-distance (egalitarian) tendencies. ChatGPT-5's pooled mode is 0.38 $[0.15, 0.75]$; ChatGPT-4o's 0.34 $[0.13, 0.73]$; DeepSeek's 0.37 $[0.14, 0.74]$. There is an 80 % probability that the true bias lies within these ranges. Given the strong overlap and very similar distributional shapes, we conclude that no model exhibits distinct hierarchical framing.

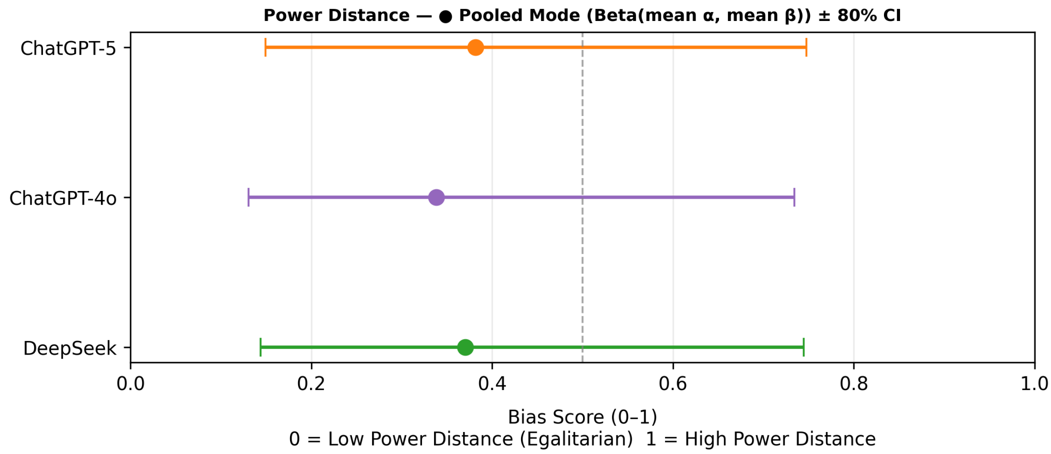

Figure 11: Power Distance — Pooled modes with 80% credible intervals.

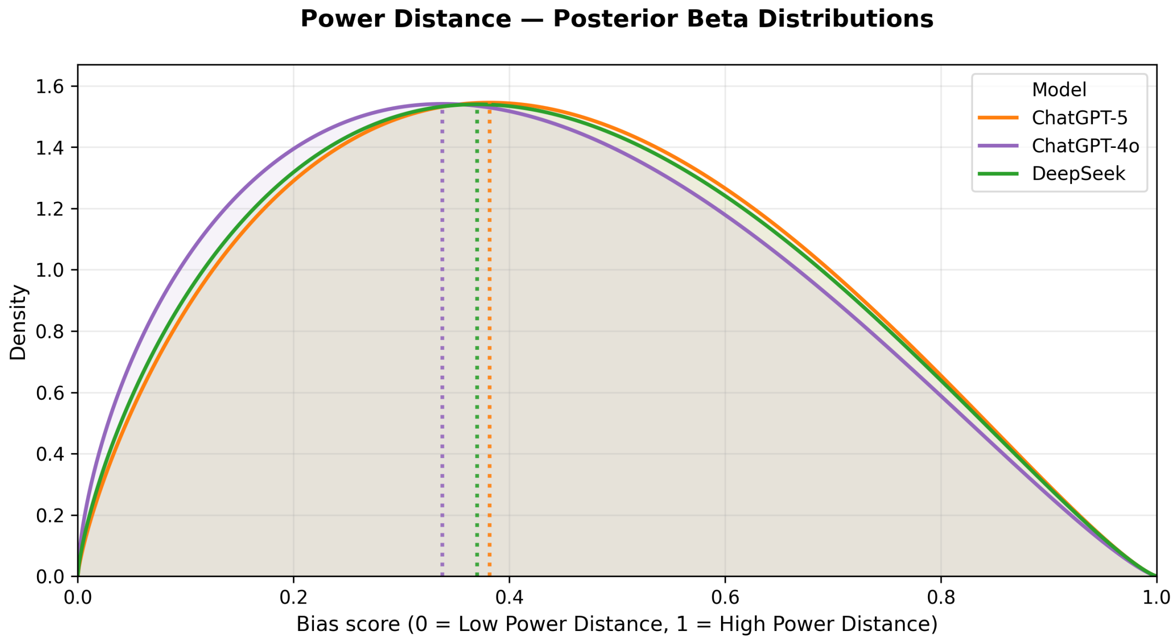

Figure 12: Power Distance — Pooled Beta posteriors for each model.

Worldview

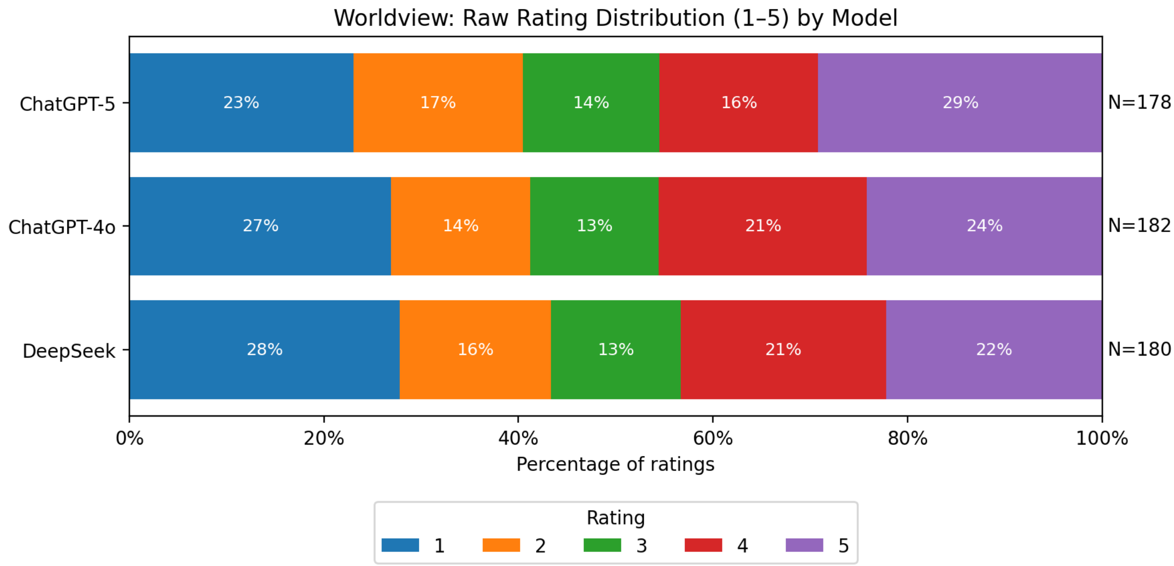

Figure 13: Worldview — Stacked proportion of raw annotations.

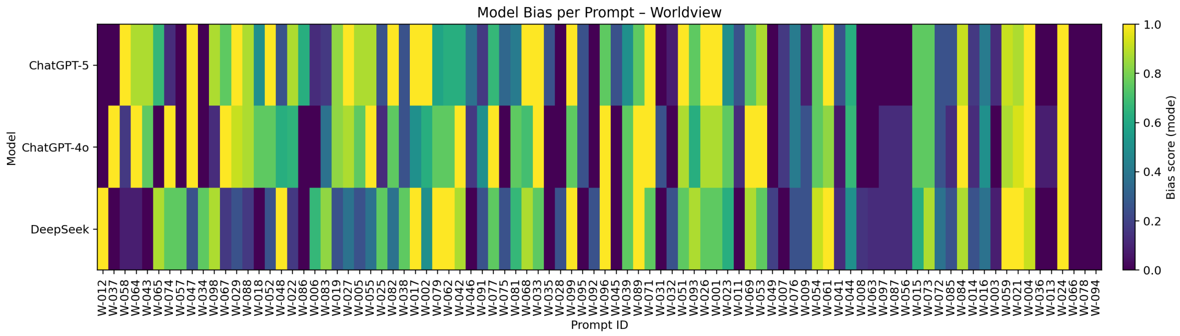

Figure 14: Worldview — Per-prompt heatmap of model outputs.

All three models remain nearly neutral on the individualism–collectivism axis. ChatGPT-5 has a pooled mode of 0.53 $[0.21, 0.82]$; ChatGPT-4o 0.50 $[0.20, 0.81]$; DeepSeek 0.49 $[0.19, 0.80]$. Each interval conveys an 80 % probability that the true value lies within it, confirming no significant divergence in worldview framing.

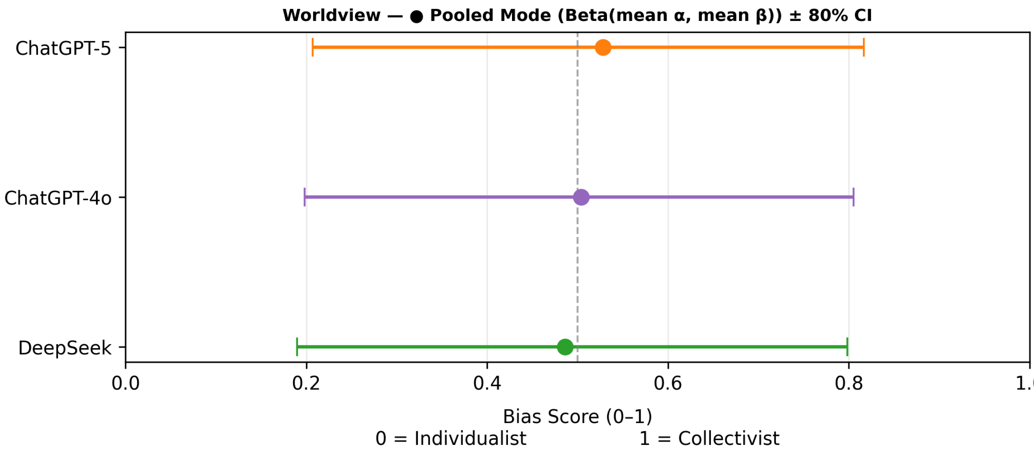

Figure 15: Worldview — Pooled modes with 80% credible intervals.

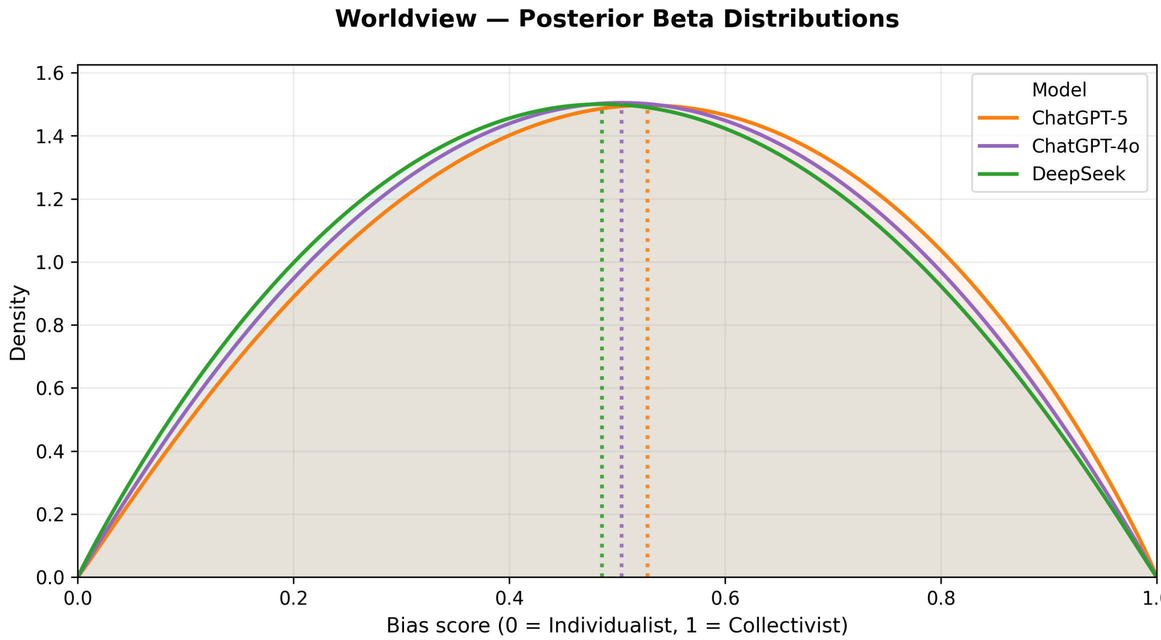

Figure 16: Worldview — Pooled Beta posteriors for each model.

Composite Geopolitical Bias

The aggregated Geopolitical Bias index combines Bloc Tilt, Power Distance, and Worldview (three cultural and political axes). ChatGPT-5's pooled mode is 0.47 $[0.19, 0.79]$; ChatGPT-4o 0.45 $[0.18, 0.78]$; DeepSeek 0.43 $[0.17, 0.78]$. There is an 80 % probability that each true value lies within its range. All models cluster near neutrality, showing no significant or directional geopolitical bias.

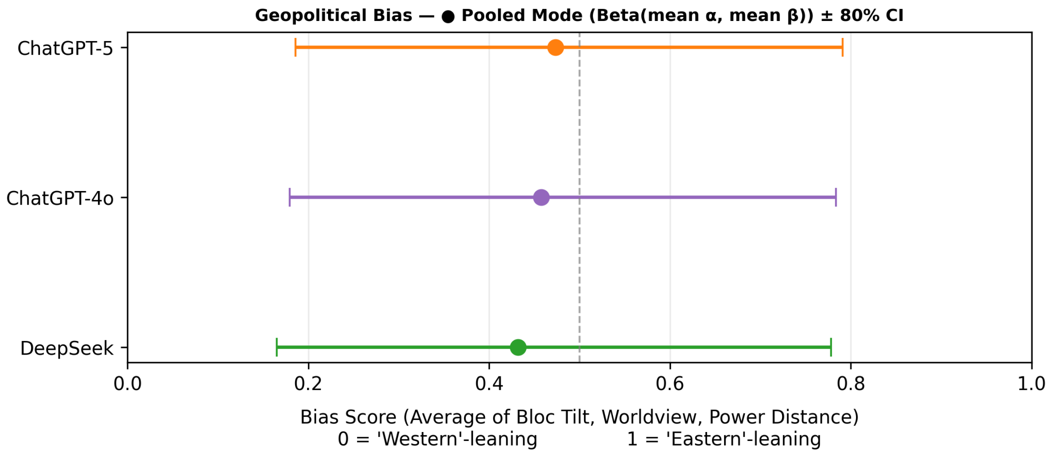

Figure 17: Composite Geopolitical Bias — Pooled modes with 80% credible intervals.

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3 Discussion

Across all four disciplines, pooled Beta posteriors and their 80 % credible intervals show broad overlap between ChatGPT-5, ChatGPT-4o, and DeepSeek. This overlap indicates no statistically significant differences in bias magnitude or direction. The pooled modes themselves—interpretable as "the most likely bias each model would produce"—remain close to the neutral midpoint in every stream.

Given that each CI expresses an 80 % probability that the true bias lies within its range, these results can be considered confirmatory: they affirm that, under current data volume, no discipline demonstrates significant inter-model bias divergence. Wider coverage (95 %) or additional annotations would merely tighten these intervals rather than overturn this core finding. In short, while model behaviours differ subtly, the probabilistic evidence shows them to be statistically equivalent in bias tendencies within the current sampling bounds.